

Ayesha Binte Mostofa

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RESEARCH INTEREST

My research interests include distributed systems, usable security, and agentic AI, and. My work focuses on understanding and addressing security and privacy challenges in software systems and online platforms. I pursue two complementary directions: (1) developing scalable and robust methods for securing large-scale systems, including AI-assisted approaches for analyzing and enforcing security policies, and (2) understanding human-centered security and privacy challenges and developing user-centered mechanisms that support safer decision-making on online platforms.

Keywords: Cloud and Network Systems Security and Privacy, Usable Security, Agentic AI

EDUCATION

University of Massachusetts Amherst (UMass Amherst)

MS/Ph.D. in Computer Science

Advisor : [Dr. Pubali Datta](#)

Sep 2025 – Present

Amherst, MA, USA

Bangladesh University of Engineering and Technology (BUET)

B.Sc.Engg. in Computer Science and Engineering

Thesis Advisor : [Dr. Anindya Iqbal](#)

April 2019 – July 2024

Dhaka, Bangladesh

RESEARCH EXPERIENCE

Security Research Group of Dr. Pubali Datta, UMass Amherst | Fall 2025-Present

- Led the development of an LLM-aided framework for automated synthesis and validation of access control policies for cloud applications.
- Led a large-scale analysis of students' scam experiences using student discussions.
- Conducted a survey-based analysis of students' scam experiences in the US, examining differences between domestic and international students.

BUET Undergrad Research | 2023-2025

- Designed prompts incorporating static program metadata (function call graphs) and natural language summaries; applied QLoRA fine-tuning to improve code review comment and code refinement generation.
Advisor: Dr. Anindya Iqbal, Dr. Toufique Ahmed
- Detected benign and malicious attack behaviors by analyzing raw log data from DARPA OPTC and DARPA TC E3 datasets using PostgreSQL and provenance graphs.
Advisor: Dr. Shohrab Hossain, Dr. Shahrear Iqbal

PUBLICATIONS

- 1. Quantifying Risk Perception and Scam Response Among International and Domestic U.S. University Students.** Alexandra Xinran Li, Elijah Bouma-Sims, Lily Klucinec, Ray Liu, **Ayesha Binte Mostofa**, Arjun Arunasalam, Lorrie Faith Cranor, Pubali Datta, Lucy Simko, Karen Sowon. *Symposium on Usable Privacy and Security (SOUPS)*, 2026.
- 2. Advancing Code Review and Code Refinement Automation Using LLMs.** **Ayesha Binte Mostofa***, Sk. Sabit Bin Mosaddek*, Asif Haider*, Toufique Ahmed, Anindya Iqbal. [arXiv Preprint](#), 2024. (*Equal contribution)
– 10th NSysS Research Poster Presentation, 2023. [Poster](#)
- 3. Understanding Synthetic Politics on Reddit: Spread and Community Reactions to Political Deepfakes.** **Ayesha Binte Mostofa** and Pubali Datta. *USENIX Security '26 Poster*.

TALKS & SERVICE

Short Talk | Annual Boston Security Usable Research Day, 2026 at Tufts University

External Reviewer | **IEEE TDSC 2026**

Poster Presentation & Volunteer | New England Security Day 2026, UMass Amherst

IEEE Computer Society BUET Student Branch Chapter, 2021 – 2023

- Publicity Committee Coordinator (2022 – 2023); Media Committee Executive (2021 – 2022)

HONORS & AWARDS

USENIX Security Symposium Travel Grant, \$825 Award, 2026

GREPSEC VII Workshop Participant, Selected as 1 of 48 participants internationally; \$1175 Travel Grant Award (co-located with USENIX Security '26)

David W. Stemple scholarship in computing 2026, \$2300 Award, UMass Amherst

RISE-BUET Internal Student Research Grant for Undergraduate Thesis, 2023

Dean's List Award, BUET, Level 3 & Level 4, 2023–2024

Deep Learning Enigma 1.0 Finalists, 2024, Object Detection Contest for Autonomous Vehicles

Merit Scholarships, 2017–2024, granted for being top 1% student in HSC and SSC level

WORK EXPERIENCE

Research Assistant , UMass Amherst	Summer 2026, Fall 2026
Teaching Assistant , UMass Amherst - Introduction to Computation (COMPSCI 250) - Programming Methodology (COMPSCI 220)	Fall 2025, Spring 2026
Full-time Lecturer <i>Department of CSE, Presidency University; [Profile]</i> <i>Department of CSE, Canadian University of Bangladesh</i>	Dhaka, Bangladesh January 2025 – July 2025 July 2024 – January 2025
Machine Learning Intern (Part-time) <i>Red.Digital Limited</i>	May 2023 – June 2023 Remote

ACADEMIC PROJECTS

LLMComp: Prompt Compression for Robust Defense Against LLM Jailbreaks — 2025	<i>LLM, Few-Shot, Jailbreak</i>
IaC-Eval Replication and Extension Study — 2025	<i>IAC, LLM, Few-shot, CoT</i>
Hardware : ALU, FPA, MIPS Implementation — 2022	<i>C++, Logisim, Assembly</i> Source Code
MISP Tool, Cryptography and Malware Analysis — 2023	<i>Python, Docker, Azure Cloud</i> Source Code — Youtube Demo
C Compiler — 2022	<i>Lex, Yacc, Assembly, C</i> Source Code
Implementing functionalities of OS — 2023	<i>C, Bash</i> Source Code
TCP CUBIC-FIT, Packet Tracer and Network Simulator — 2022	<i>Java, NS2</i> Source Code

TECHNICAL SKILLS

Languages	: C/C++, Python, Java, Javascript, Assembly, Bison/Flex, Bash, MySQL, LaTeX
Frameworks	: React.js, Node.js, SpringBoot, Oracle, PostgreSQL, Docker, NS2, xv6, Git, Wireshark
Libraries	: NumPy, Keras, Matplotlib, OpenCV, OpenGL, Pandas, Scikit Learn
Platform	: Linux, Windows, MacOS

REFERENCE

Dr. Pubali Datta <i>Assistant Professor, Computer Science</i> PhD Advisor	University of Massachusetts, Amherst <i>pdatta@umass.edu</i>
Dr. Anindya Iqbal <i>Professor, Computer Science and Engineering</i> Undergrad Thesis Advisor	Bangladesh University of Engineering and Technology <i>anindya@cse.buet.ac.bd</i>